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«Брестский государственный технический университет»
Кафедра ИИТ

Лабораторная работа №8
По дисциплине: «Аппаратно-программное обеспечения ЭВМ и сетей»

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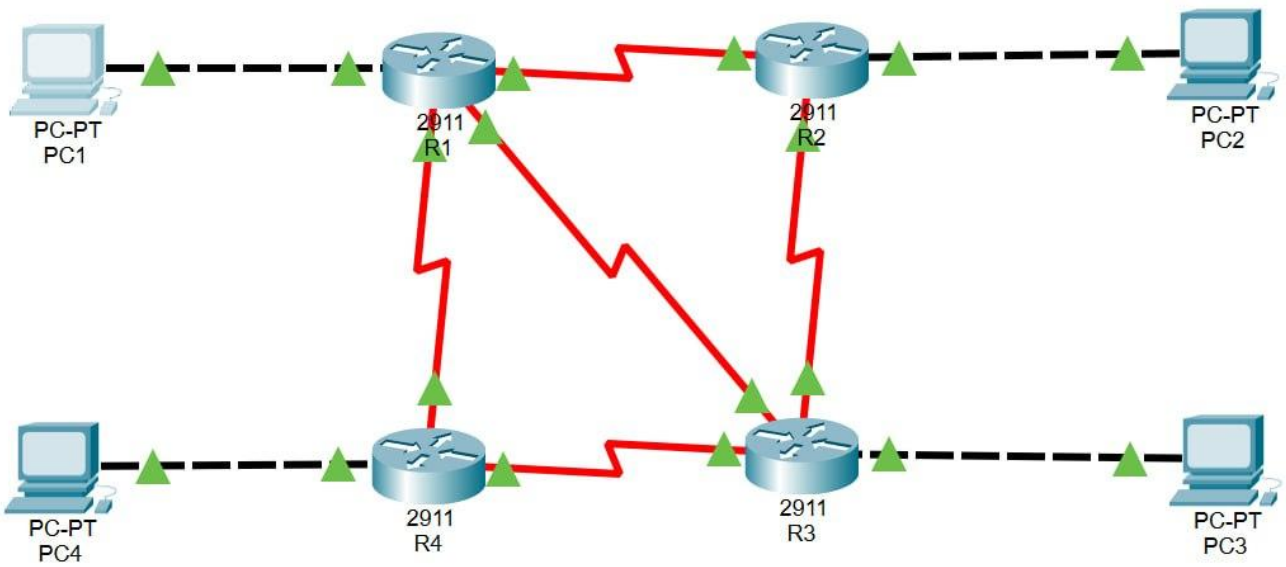
Брест 2026

Цель работы: Настроить динамическую маршрутизацию с помощью протокола RIP на устройствах R1, R2, R3; обеспечить возможность взаимодействия конечных устройств PC1, PC2, PC3 между собой.

Вариант №6

Ход работы

Device	Interface	IP Address	Mask	Default Gateway
R1	Gi0/0	192.168.10.1	255.255.255.0	N/A
	Se0/0/0	10.0.12.1	255.255.255.252	N/A
	Se0/0/1	10.0.13.1	255.255.255.252	N/A
	Se0/1/0	10.0.14.1	255.255.255.252	N/A
R2	Gi0/0	192.168.20.1	255.255.255.0	N/A
	Se0/0/0	10.0.12.2	255.255.255.252	N/A
	Se0/0/1	10.0.23.1	255.255.255.252	N/A
R3	Gi0/0	192.162.3.18	255.255.255.0	N/A
	Se0/0/0	10.0.13.2	255.255.255.252	N/A
	Se0/0/1	10.0.23.2	255.255.255.252	N/A
	Se0/1/0	10.0.34.1	255.255.255.252	N/A
R4	Fa0/0	192.168.4.18	255.255.255.0	N/A
	Gi0/0/0	10.0.14.2	255.255.255.252	N/A
	Se0/0/1	10.0.34.2	255.255.255.252	N/A
PC1	N/A	192.168.10.10	255.255.255.0	192.168.10.1
PC2	N/A	192.168.20.10	255.255.255.0	192.168.20.1
PC3	N/A	192.168.30.10	255.255.255.0	192.168.30.1
PC4	N/A	192.168.40.10	255.255.255.0	192.168.40.1



R1

```
enable
configure terminal

interface GigabitEthernet0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface Serial0/0/0
ip address 10.0.12.1 255.255.255.252
clock rate 64000
no shutdown
exit

interface Serial0/0/1
ip address 10.0.13.1 255.255.255.252
clock rate 64000
no shutdown
exit

interface Serial0/1/0
ip address 10.0.14.1 255.255.255.252
clock rate 64000
no shutdown
exit

end
write memory

enable
configure terminal
router rip
version 2
no auto-summary
network 10.0.0.0
network 192.168.10.0
passive-interface GigabitEthernet0/0
end
write memory
```

R2

```
enable
configure terminal

hostname R2

interface GigabitEthernet0/0
ip address 192.168.20.1 255.255.255.0
no shutdown
exit

interface Serial0/0/0
ip address 10.0.12.2 255.255.255.252
clock rate 64000
no shutdown
exit

interface Serial0/0/1
ip address 10.0.23.1 255.255.255.252
clock rate 64000
no shutdown
exit

end
write memory
```

```
enable
configure terminal
router rip
version 2
no auto-summary
network 10.0.0.0
network 192.168.20.0
passive-interface GigabitEthernet0/0
end
write memory
```

R3

```
enable
configure terminal

hostname R3

interface GigabitEthernet0/0
ip address 192.168.30.1 255.255.255.0
no shutdown
exit

interface Serial0/0/0
ip address 10.0.13.2 255.255.255.252
no shutdown
exit

interface Serial0/0/1
ip address 10.0.23.2 255.255.255.252
no shutdown
exit

interface Serial0/1/0
ip address 10.0.34.1 255.255.255.252
clock rate 64000
no shutdown
exit

end
write memory

enable
configure terminal
router rip
version 2
no auto-summary
network 10.0.0.0
network 192.168.30.0
passive-interface GigabitEthernet0/0
end
write memory
```

R4

```
enable
configure terminal

hostname R3

interface GigabitEthernet0/0
ip address 192.168.30.1 255.255.255.0
no shutdown
exit

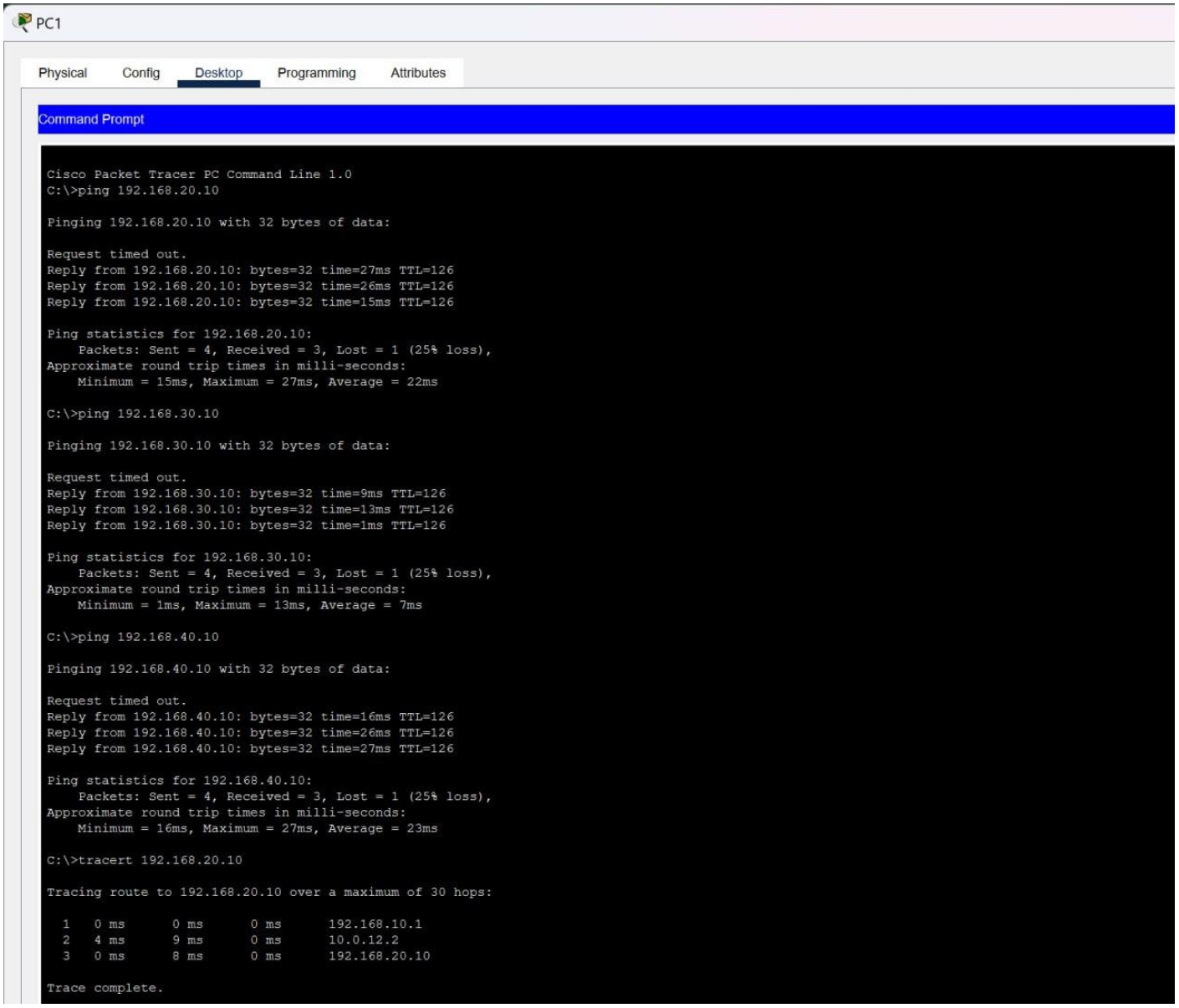
interface Serial0/0/0
ip address 10.0.13.2 255.255.255.252
no shutdown
exit
```

```
interface Serial0/0/1
ip address 10.0.23.2 255.255.255.252
no shutdown
exit

interface Serial0/1/0
ip address 10.0.34.1 255.255.255.252
clock rate 64000
no shutdown
exit

end
write memory

enable
configure terminal
router rip
version 2
no auto-summary
network 10.0.0.0
network 192.168.30.0
passive-interface GigabitEthernet0/0
end
write memory
```



Вывод: я настроил динамическую маршрутизацию с помощью протокола RIP на устройствах R1, R2, R3; обеспечила возможность взаимодействия конечных устройств PC1, PC2, PC3 между собой.